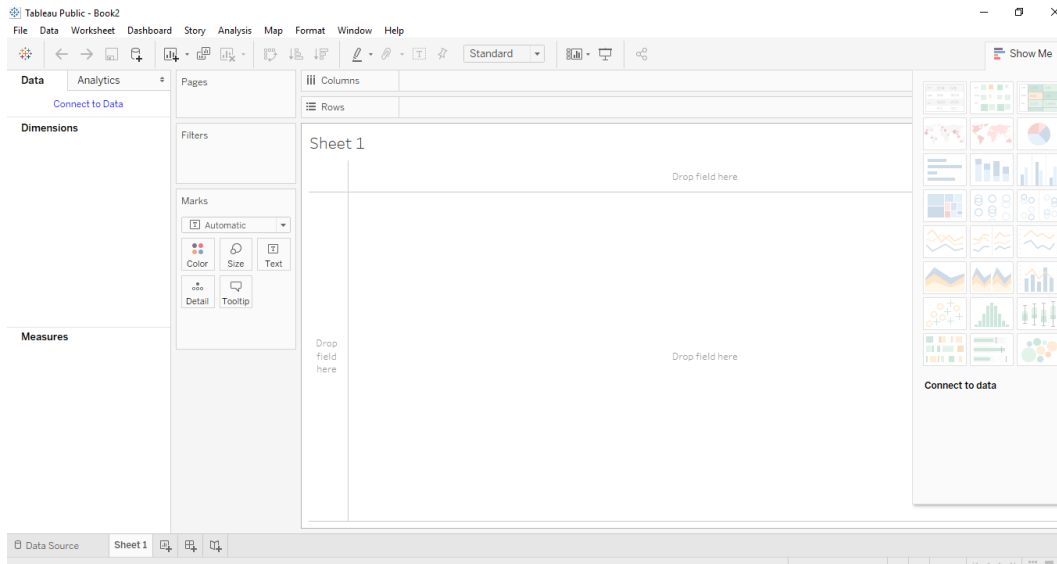


Data Visualisation SPEC9995

Lab 1 Tableau Getting Started

Main screen

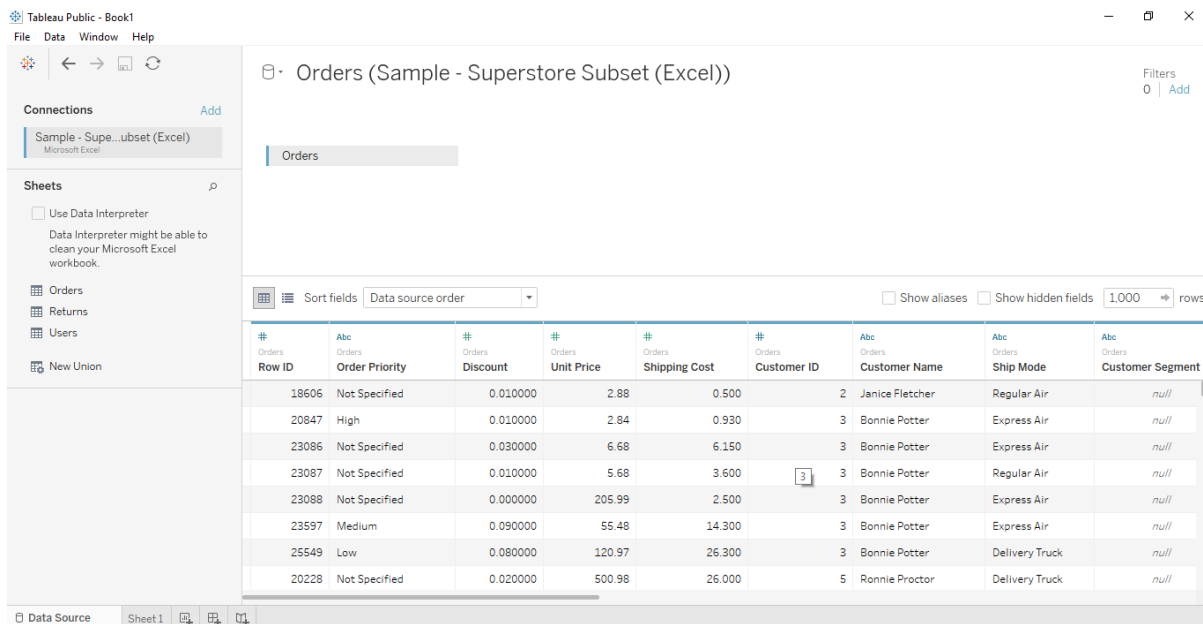


Connecting to Data and getting started

Open the sample superstore Excel file. Tableau is a very user-friendly software that works mainly on drag and drop. Select the variable or action you want to perform and drop it in the chart area.

In the superstore file we can find 3 main tables: Orders, Returns and Users.

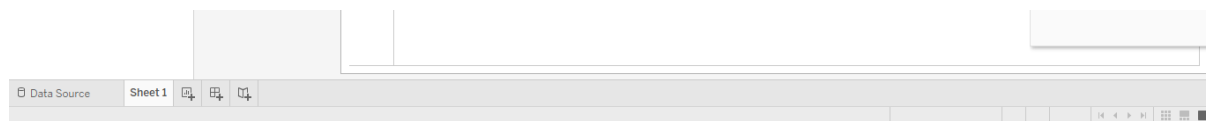
The Datasource Tab contains the information about the data available and allows for raw data manipulation. Drag the Orders table to the data sheet area.



You can look at the Metadata for field names and data structure. You can edit datatypes, names etc..

Field Name	Table	Remote Field Name
# Row ID	Orders	Row ID
Abc Order Priority	Orders	Order Priority
# Discount	Orders	Discount
# Unit Price	Orders	Unit Price
# Shipping Cost	Orders	Shipping Cost
# Customer ID	Orders	Customer ID
Abc Customer Name	Orders	Customer Name
Abc Ship Mode	Orders	Ship Mode
Abc Customer Segment	Orders	Customer Segment
Abc Product Category	Orders	Product Category

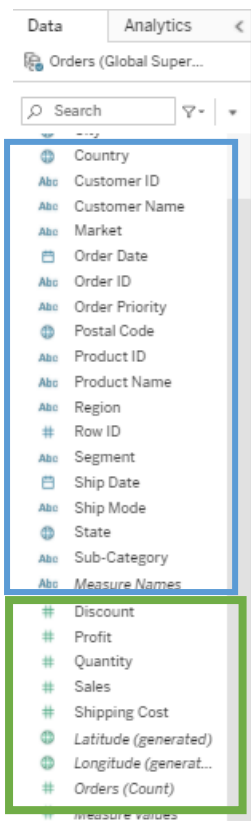
To go to the development canvas in Tableau, click on Sheet1 at the bottom of the page.



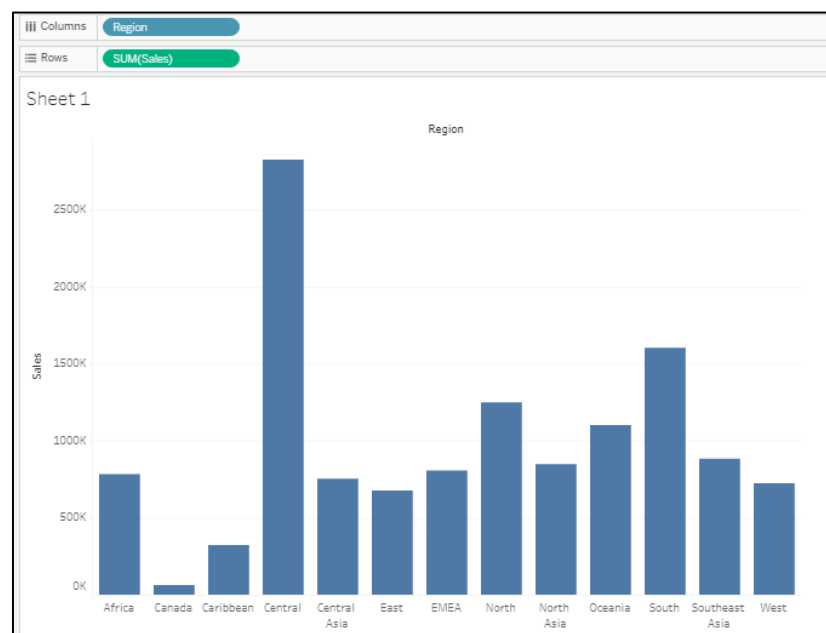
Exercise 1. Creating a column chart displaying sales per region

To include a variable in a chart, you can drag it into the appropriate area (Columns/ Rows).

Variables are split into two groups: discrete dimensions for categorical data and measurement for continuous data. Discrete variables will be displayed in blue and will create labels in the chart, whereas continuous variables will be displayed in green and will create axis in the chart.



We can create a basic chart showing the sales per region by dragging Sales from the Data panel into the Rows box and Region into the columns box.

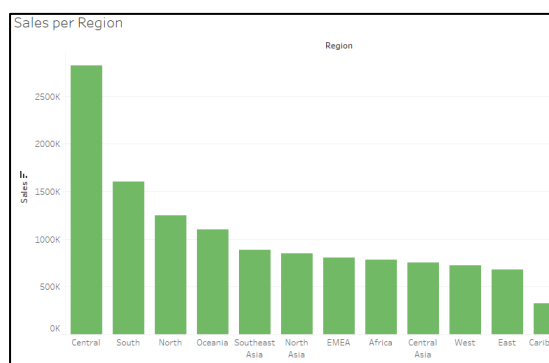
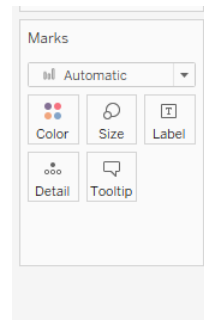


Customizing our chart

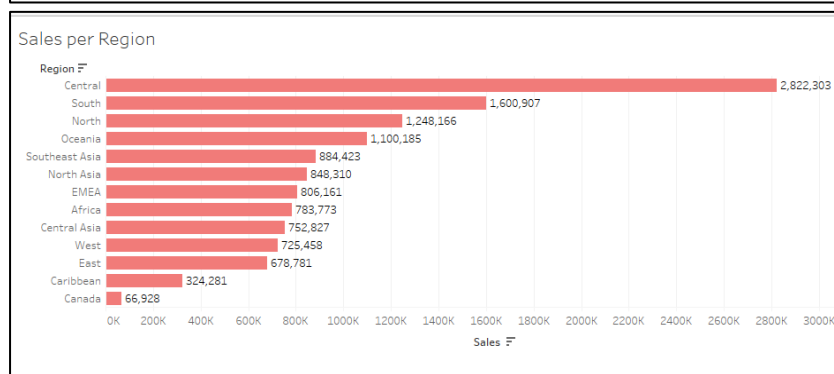
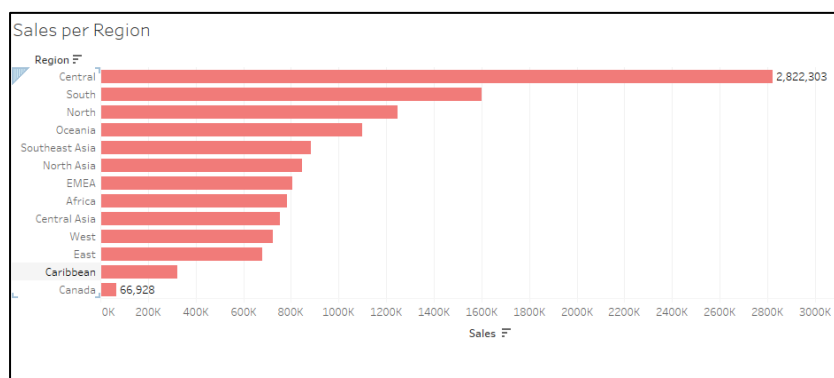
Order your columns from highest to lowest using the sort button on the toolbar (1). We can also swap rows and columns (2) or sort in ascending order (3).



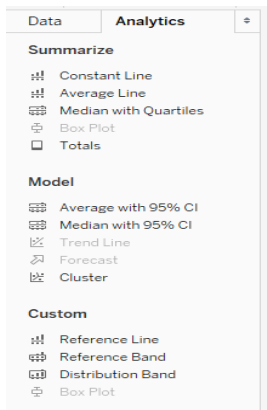
Colours and titles can also be customized. Use the Color menu within the Mark panel to change the colour of the bars. A variable can also be dragged into the colour menu to customize colours based on that variable, for example, we could arrange a different colour per region or highlight a particular value of interest.



Adding mark labels to each data point or to specific ones can be done using the Label menu within the Marks panel. Two examples are shown below.

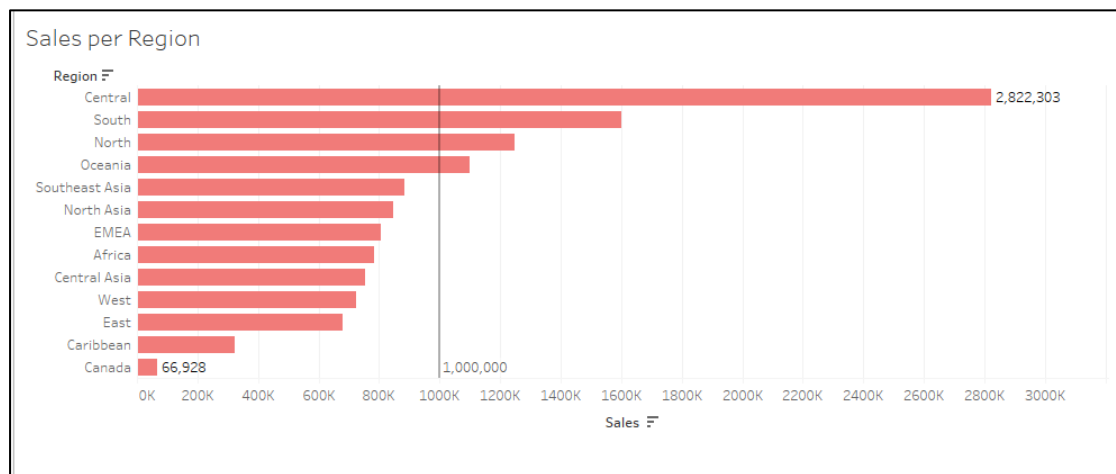


Exercise 2. Adding reference values to the chart



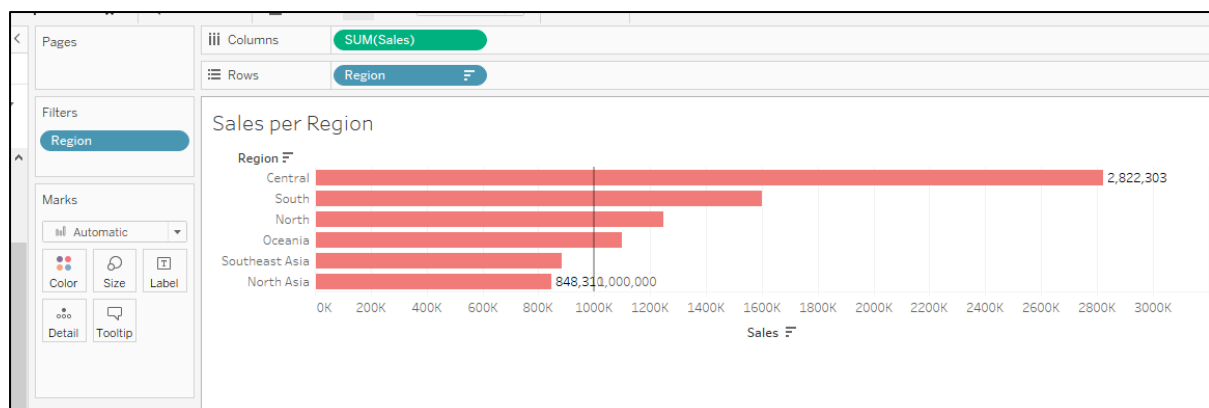
From the Analytics tab, we can add different summarization, modelling or custom reference values to our chart.

Add a constant value line at 1 million as reference line. HINT: Drag and drop the Constant Line feature into the canvas and set the value. Similarly, average values and other statistical measures can be added.



Exercise 3. Filtering the Data.

A number of different filters can be used in Tableau. To create a filter, drag the particular dimension or measure that you want to use to filter into the Filter box and customize it to show the top 6 regions in terms of sales.



Exercise 4. Let's answer some questions

- A- How does Profit perform per country? Use a map to show the worldwide distribution of profit.
- B- How does Quantity compare to Profit?
- C- How does Quantity perform per product category?
- D- Show the sales per year and quarter.

Save your Tableau file to Tableau Public.

References

Sinha, C. (2017) Tableau 10 for beginners. Ohio Computer Academy.